

Cargo Operations

Cargo Space in Bulk Terminals

Exercise 1

Bulk cargo includes all loose dry goods.

Bulk cargo also includes oil and gases.

Coal, ore, grain, salt and fertilizers are examples of bulk cargo.

Coal is loaded by belt conveyors or crane grabs.

Coal is exported from Poland.

Coal is stored in the yard.

Oil is transported as liquid cargo.

Oil is loaded by pipelines.

Oil is imported to Poland.

Crude oil is stored in the tanks.

Exercise 2

	<t>	<d>	<id>
Coal is usually loaded by belt conveyors.			X
It can be discharged by grabs.		X	
Grain is stored in s silo.		X	
Oil is transported by pipelines.			X
Bulk cargo is trimmed by bobcats.		X	
Holds are washed before loading cargo.	X		

Cargo Space in Container Terminals

Exercise 1

steel – still

feet – fit

ship – **sheep**

meal - mill

chip – **cheap**

eat – it

keel – kill

heat - hit

deep – dip

feel – fill

leak – lick

jeep – jib

beach – bitch

Exercise 2

bar	car	far
click	creek	clip
cube	tube	tub
tank	rank	tan
sun	fun	run
space	race	place
foot	food	ford
twist	tweed	twin
lock	look	lift
block	clock	rock
bridge	fridge	ridge

Exercise 3

a twistlock	a bridge fitting	
a lashing chain	lashing bars	You can see what happens when containers are not lashed properly

Exercise 4

Lock the twistlocks	twistlocks locked
Load reefer containers	reefer containers loaded
Check the twistlocks	twistlocks checked
Fasten the lines	lines fastened
Lash the cargo	cargo lashed
Replace lashing bars	lashing bars replaced
Collect lashing chains	lashing chains collected
Put bridge fittings aside	bridge fittings put aside
Unlock the twistlocks	twistlocks unlocked
Fasten the lashing wires	lashing wires fastened
Fit the crossbars	crossbars fitted

Liquid Cargo Operations

Exercise 1

Answer Key not available

Exercise 2

a jetty	has been built from the shore into the water
hazardous cargo	may cause harm to the environment
ullage	is the space above the liquid in a tank
crossover valves	separate the main lines from each other
master valves	separate tanks in a fore-and-aft direction
drop lines	lead vertically from the deck lines to the tank lines
stripping lines	are used in tank washing and ballast operations
oil product	is refined from crude oil

Exercise 3

A system of pipes connects cargo tanks to **<manifolds>**.

Portable **<drip trays>** are placed under all ship and shore connections.

The **<crossover>** valves separate the main lines from each other.

Wing tanks are on port and **<starboard>** sides of the ship.

The ship should have emergency towing-off **<wires>** ready.

The tanks are **<topped-off>**.

There are joints, **<bends>** and valves in the pipelines.

The cargo is pumped on board by shore **<pumps>**.

Exercise 4

1. <**Moor/moor**> the ship!
I <**have moored**> the ship
2. <**Connect/connect**> the loading arms!
I <**have connected**> the loading arms.
3. <**Check/check**> the valves!
I <**have checked**> the valves.
4. <**Open/open**> tanker valves!
I <**have opened**> the tanker valves.
5. <**Start/start**> the pumps!
I <**have started**> the pumps.
6. <**Keep/keep**> a close watch for oil escaping!
I <**have kept**> a close watch for oil escaping.
7. <**Stop/stop**> loading! (St)
I <**have stopped**> loading.
8. <**Close/close**> the valves!
I <**have closed**> the valves.
9. <**Drain/drain**> the cargo deck lines!
I <**have drained**> the cargo deck lines.
10. <**Disconnect/disconnect**> the loading arms!
I <**have disconnected**> the loading arms.
11. <**Disconnect/disconnect**> the earthing cable!
I <**have disconnected**> the earthing cable.
12. <**Blank/blank**> cargo manifolds!
I <**have blanked**> the cargo manifolds.

Exercise 5

moor	cast off
correct	wrong
port	starboard
open	closed
empty	full
load	discharge
manually	automatically
possible	impossible
connect	disconnect
fore	aft

Exercise 6

New tankers have a double hull which gives **<protection>** to the cargo.

Manifolds **<connect>** deck pipelines with shore hoses or loading arms.

A crew member is standing at the manifolds, in the case of a leaking **<connection>** or any other problem.

The main pipelines **<connect>** with cargo pumps and the deck pipelines.

Tankers **<carry>** liquid bulk cargo in cargo tanks.

The master of a tanker is responsible for the safe and secure **<mooring>** of the ship.

Before **<pumping>** begins, all terminal and tanker valves in the transfer system are opened.

<Loading> starts at a low rate to a single tank.

The crew member is then ready to instruct the shore to stop **<loading>**.

At the end of **<loading>** operation the cargo tanks are topped-up.

The **<loading>** is finished and all valves in the cargo system are closed.

Oil terminals are situated in locations where tankers can **<load>** or discharge their cargo.

Dangerous Goods

Exercise 1

When carrying goods that are considered to be **dangerous**, you have to **fill in** a special **form**.

In that form, you have to write down the name of the **shipper**, the **consignee** and the **carrier**.

You also have to give **details** about your company, or signatory **organization**.

In the declaration, you state that the packing of goods has been done according to **IMGD Code** rules.

Other data to be included are: the ship's name and **voyage**, the port of **loading**, the port of **discharge**.

The form usually ends with a descriptive list of the cargo, including **special marks**, number and kind of **packages**.

Exercise 2

Flammable gas	Flammable solids	Spontaneously combustible goods	Dangerous when wet
Explosive goods	Non-flammable compressed gas	Toxic gas	Miscellaneous dangerous goods or articles

Exercise 3

The International Maritime Dangerous Goods (IMDG) Code	deals with provisions about carrying special goods.
That may be either dangerous, or are marine pollutants,	or both.
Cargoes are divided into different classes,	according to the kind of hazard.
Some classes may even be subdivided	into other smaller groups.
Goods are grouped according	to the most predominant hazard.
The first three groups are	explosives, gases and flammable liquids.
Flammable solids, oxidizing substances and organic peroxides,	and toxic and infectious substances form the three following groups.
They are followed by	radioactive material and corrosive substances.
Finally, other dangerous articles and substances	form a group of their own.

Roro Vessels

Exercise 1

You can enter a Ro-Ro vessel through a **stern ramp**.

This is the **bow** of the vessel.

Berthing operations are performed from the **bridge wings**.

The **service side gates** are used for maintenance operations.

Smoke is evacuated through the vessel's **funnel**.

The **aerial** is used to communicate.

The **lifeboats** are used in emergency situations.

Saloons and cabins are on the **passenger deck**.

The **bulbous bow** helps the ship sail better.

Most passengers go onboard through the **entrance door**.

Exercise 2



This is a fast-ferry catamaran Ro-Ro vessel

This is a traditional Ro-Ro Vessel



This is a conventional ferry

Passengers may get onboard through a gangway or a gangplank

Exercise 3

1. What is the **safety load** of no. 5 hold?
2. How many **cubic metres** of cargo space are required?
3. What is the **deadweight** of the vessel?
4. How many **20'/40' containers** will the vessel load?
5. Are the **cranes/derricks** operational?